PROJECT

REPORT

Analysis of Crop production due to rainfall in India

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Submitted To-

Dr. Sachin Tawra Sir

DECLARATION

We, certify that this project is our own work, based on our personal study and/or research and that we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. We also certify that this project has not previously been submitted for assessment in any academic capacity, and that we have not copied in part or whole or otherwise plagiarised the work of other persons. We confirm that we have identified and declared all possible conflicts that we may have.

CERTIFICATION

This is to certify that this group of studends has successfully completed the Analysis of Crop production due to rainfall in India project under the guidance of Mr. Sachin Tawra at Atma Ram Sanatan Dharma College, University of Delhi. This project was undertaken as a part of Data visualization and techniques during the 2024-25 session.

Project Details:

➢ Project Title: *Analysis of Crop production due to rainfall in India*

➢ Supervisor/Instructor: Mr. Sachin Tawra

➢ Course: Data engineering and Analytics

During the course of this project, this group demonstrated exceptional skills, commitment, and dedication to the successful completion of the assigned tasks. The project involved data analysis of Rainfall pattern, and the outcomes reflect the participant's proficiency in applying theoretical knowledge to practical scenarios.

Mr. Sachin Tawra has evaluated the project work and confirms that it meets the required standards for completion. The participant has shown a high level of understanding, creativity, and diligence in executing the project tasks.

ACKNOWLEDGEMENT

I would like to thank all those people who have contributed both directly and indirectly to my work especially all the teachers and students in schools who gave freely their time to complete the questionnaires and answer interview questions. Others who took part in the discussion forum deserve mention as well.

Special mention goes to my supervisor Dr. Sachin Tawra for his dedication, scholarly comments and suggestions that formed an invaluable part of my work. I cannot fail to mention him for his invaluable advice, encouragement and support.

I am also thankful to Mr. Gyantosh Kumar Jha, Principal Atma Ram Sanatan Dharma College, University of Delhi, Delhi for extending his help and guidance in conducting all the psychological tests in the school. If it was not for his support this work would have lacked credibility.

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INSTALLATION

Matplotlib :

In the Matplotlib homepage, click on the download link

In the download page, click

matplotlib-1.3.0.win32-py2.7.exe

Download and install

Under c:\Python26\Lib\Site-packages, 2 new folders matplotlib

mpl_toolkits

Write in Command Prompt

Pandas py –m pip install pandas

Numpy py –m pip install numpy

Folium -m pip install folium

DOCUMENTATION

1. CROP YIELD DATASET:

https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset



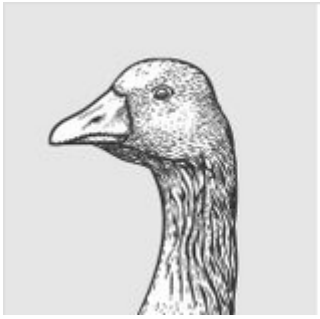
Agricultural Crop Yield in Indian States Dataset

Crop yields of Indian States and UTs from year 1997-2020

k kaggle.com

2. RAINFALL DATASET:

https://www.kaggle.com/code/darsh79/starter-rainfall-in-india-99bfc809-4/notebook



Starter: Rainfall in India 99bfc809-4

Explore and run machine learning code with Kaggle Notebooks | Using data from Rainfall in India

k Kaggle / Feb 11, 2019

INTRODUCTION

Agriculture is the backbone of the Indian economy. It contributes significantly to the nation's GDP and provides employment to more than half of India's population, either directly or indirectly. India is among the top producers of various agricultural commodities such as rice, wheat, pulses, sugarcane, and spices. However, Indian agriculture is largely dependent on climatic conditions, particularly rainfall, making it vulnerable to changes in weather patterns.



The dependence on monsoons is especially high because nearly 50% of the total cultivated area in India is rainfed, with limited or no access to reliable irrigation systems. This makes rainfall not just a weather event, but a critical factor determining food security, farmer income, and overall rural development in the country.

Rainfall in India: Distribution and Characteristics

India receives most of its rainfall during the Southwest Monsoon season (June to September). The Indian monsoon is one of the most complex weather systems in the world, and its onset, progression, and withdrawal have a profound impact on the agricultural calendar.



Key features of rainfall in India include:

Seasonality: Over 75% of the annual rainfall occurs during just four months.

Spatial variability: Some regions like the Western Ghats, northeastern states, and coastal areas receive over 2000 mm of rainfall annually, while others like Rajasthan and parts of Gujarat receive less than 500 mm.

Inter-annual variation: Rainfall levels can fluctuate significantly from year to year, leading to droughts or floods.

Due to these factors, predicting and managing rainfall is a major challenge for agricultural planning in India.

Regional Impacts of Rainfall on Crop Production

Different regions in India experience different impacts of rainfall on agriculture:

Punjab and Haryana: These states have developed irrigation systems, so crop production is relatively less sensitive to rainfall variability.



Maharashtra, Madhya Pradesh, Odisha, Chhattisgarh: Rainfed agriculture is dominant, making these states highly vulnerable to monsoon fluctuations.

Southern and Eastern states: Crops like paddy and pulses show strong correlation with rainfall variability.

In drought-prone areas like Marathwada and Bundelkhand, the lack of rainfall can lead to severe agrarian crises and even farmer suicides.



CODES

- Importing the required libraries:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from wordcloud import WordCloud # type: ignore
```

- Importing the required datasets:

```
# Load datasets
cp_df = pd.read_csv("crop_yield.csv")
rf_df = pd.read_csv("rainfall_in_india.csv")
```

- Removing unnecessary columns and filling null values and displaying the datasets:

```
# Display first few rows of each dataset
print("Crop Yield Dataset:")
print(cp_df.head())

print("\nRainfall Dataset:")
print(rf_df.head())

# Fill missing values
crop_df = cp_df.fillna(0)
rainfall_df = rf_df.fillna(0)

# Drop unnecessary columns
crop_df = crop_df.drop(columns=["Fertilizer", "Pesticide"], errors="ignore")

# Filter years 2000-2015
crop_df = crop_df[(crop_df["Crop_Year"] >= 2000) & (crop_df["Crop_Year"] <= 2015)]
rainfall_df = rainfall_df[(rainfall_df["YEAR"] >= 2000) & (rainfall_df["YEAR"] <= 2015)]
```

Printing datasets

```
==== RESTART: C:\Users\arpit\OneDrive\Desktop\abhilasha\GE PRACTICAL\test.py ===
Crop Yield Dataset:
           Crop  Crop_Year       Season  ...  Fertilizer  Pesticide        Yield
0      Arecanut       1997  Whole Year   ...   7024878.38   22882.34     0.796087
1     Arhar/Tur       1997  Kharif       ...    631643.29    2057.47     0.710435
2   Castor seed       1997  Kharif       ...     75755.32     246.76     0.238333
3      Coconut        1997  Whole Year   ...   1870661.52    6093.36  5238.051739
4  Cotton(lint)       1997  Kharif       ...    165500.63     539.09     0.420909

[5 rows x 10 columns]

Rainfall Dataset:
                SUBDIVISION  YEAR   JAN  ...  Unnamed: 19  Unnamed: 20  Unnamed: 21
0  ANDAMAN & NICOBAR ISLANDS  1901  49.2  ...          NaN          NaN          NaN
1  ANDAMAN & NICOBAR ISLANDS  1902   0.0  ...          NaN          NaN          NaN
2  ANDAMAN & NICOBAR ISLANDS  1903  12.7  ...          NaN          NaN          NaN
3  ANDAMAN & NICOBAR ISLANDS  1904   9.4  ...          NaN          NaN          NaN
4  ANDAMAN & NICOBAR ISLANDS  1905   1.3  ...          NaN          NaN          NaN

[5 rows x 22 columns]
>>
```

VISUALIZATIONS

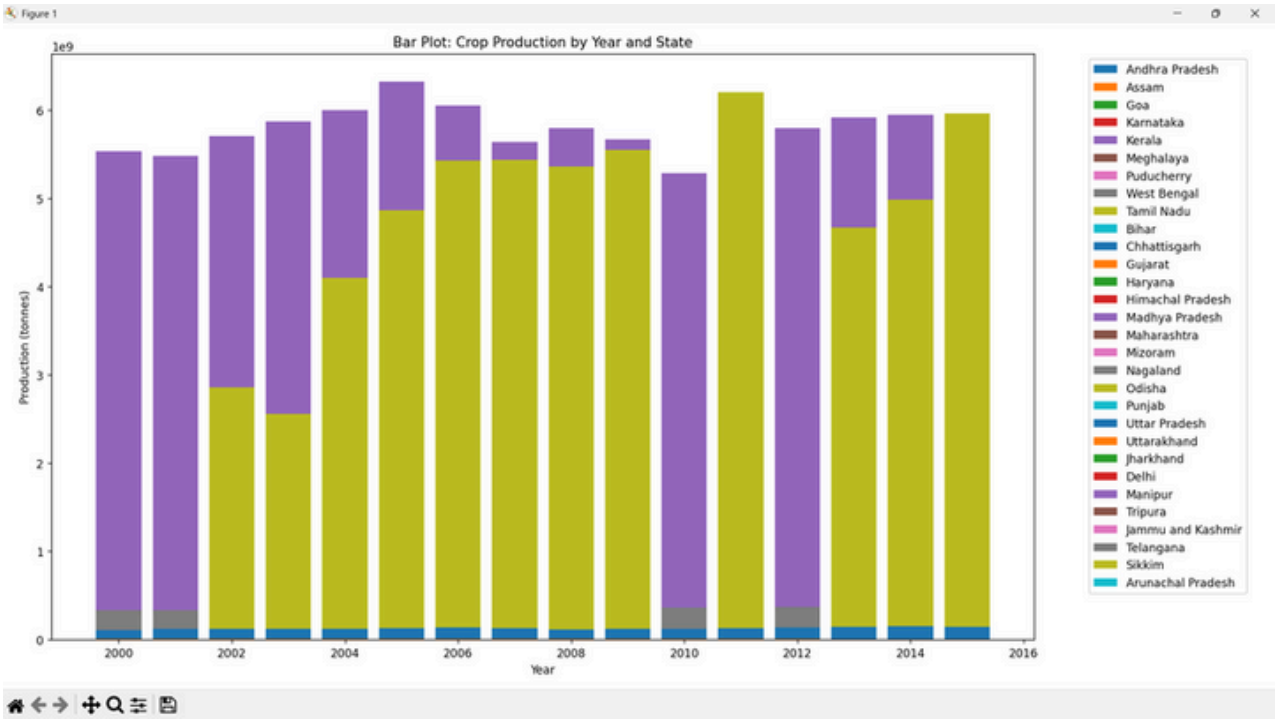
1. BAR PLOT

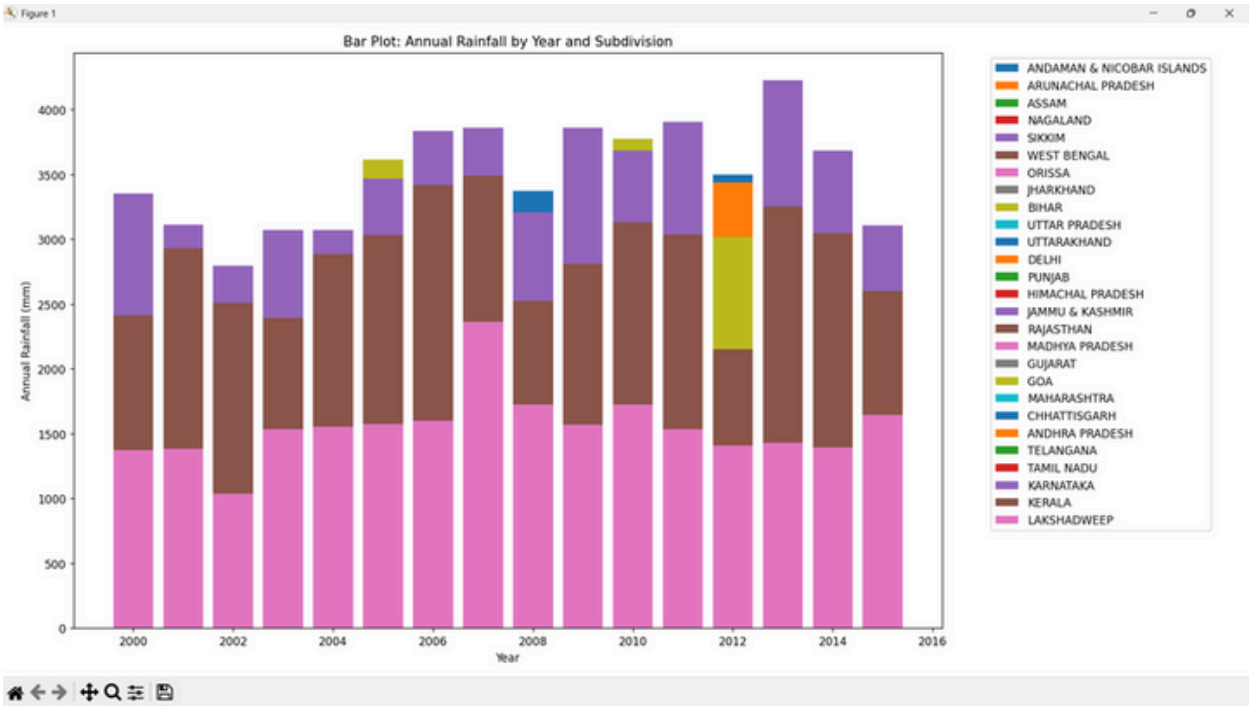
CODE:

```
# --- BAR PLOTS ---
plt.figure(figsize=(14, 8))
for state in crop_df['State'].unique():
    state_data = crop_df[crop_df['State'] == state]
    plt.bar(state_data['Crop_Year'], state_data['Production'], label=state)
plt.title("Bar Plot: Crop Production by Year and State")
plt.xlabel("Year")
plt.ylabel("Production (tonnes)")
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

plt.figure(figsize=(14, 8))
for sub in rainfall_df['SUBDIVISION'].unique():
    sub_data = rainfall_df[rainfall_df['SUBDIVISION'] == sub]
    plt.bar(sub_data['YEAR'], sub_data['ANNUAL'], label=sub)
plt.title("Bar Plot: Annual Rainfall by Year and Subdivision")
plt.xlabel("Year")
plt.ylabel("Annual Rainfall (mm)")
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

OUTPUT:





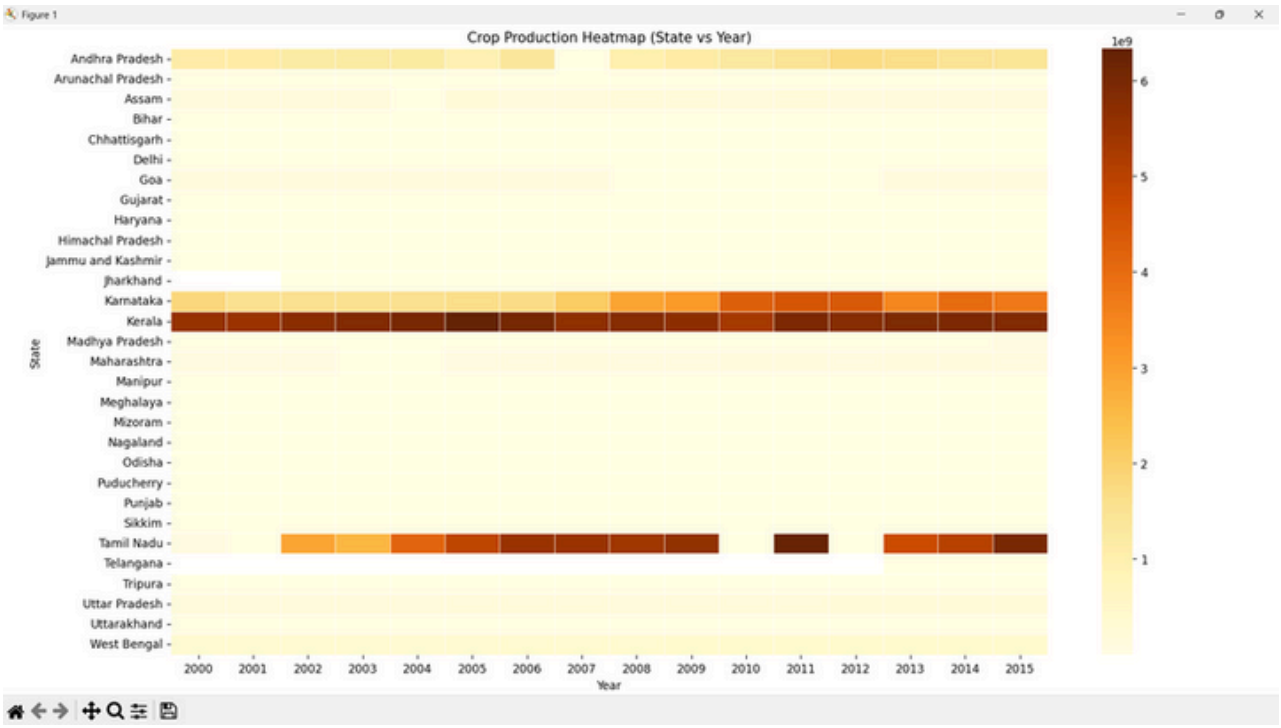
2. **HEAT MAP**

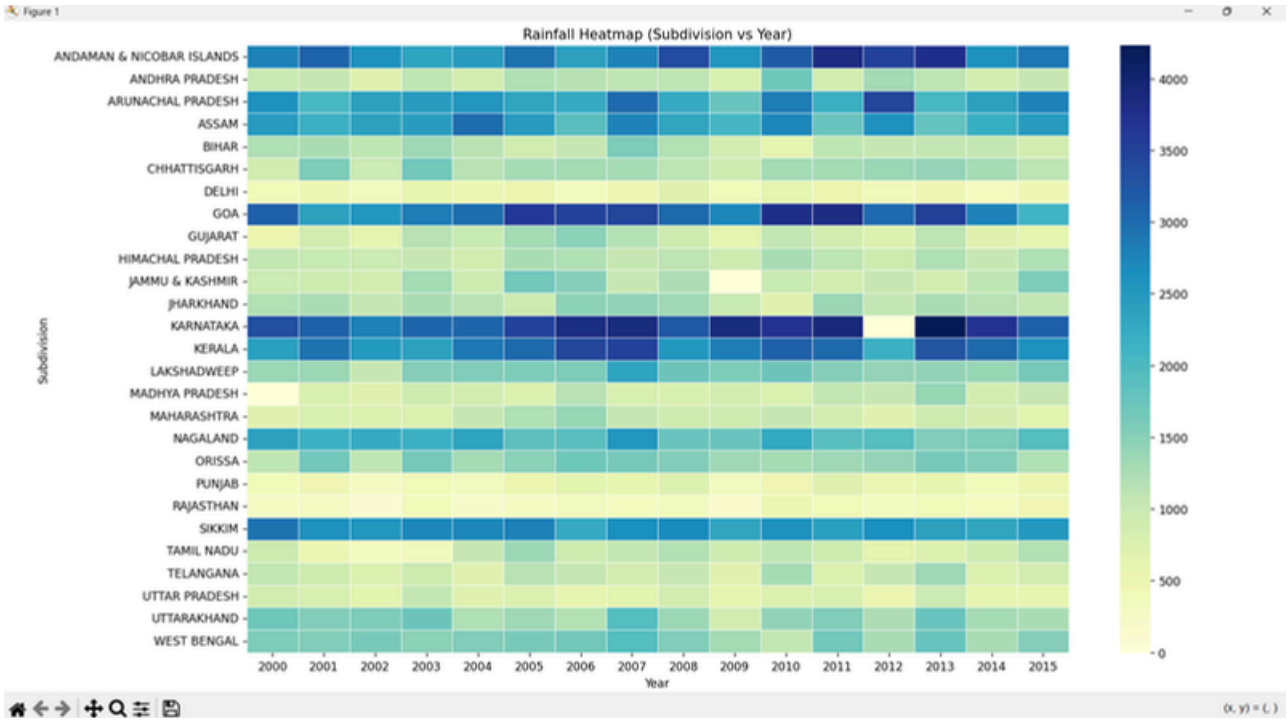
CODE:

```
# --- HEATMAPS ---
rainfall_pivot = rainfall_df.pivot_table(index="SUBDIVISION", columns="YEAR", values="ANNUAL")
plt.figure(figsize=(14, 10))
sns.heatmap(rainfall_pivot, cmap="YlGnBu", linewidths=0.5)
plt.title("Rainfall Heatmap (Subdivision vs Year)")
plt.xlabel("Year")
plt.ylabel("Subdivision")
plt.tight_layout()
plt.show()

crop_pivot = crop_df.pivot_table(index="State", columns="Crop_Year", values="Production", aggfunc="sum")
plt.figure(figsize=(14, 10))
sns.heatmap(crop_pivot, cmap="YlOrBr", linewidths=0.5)
plt.title("Crop Production Heatmap (State vs Year)")
plt.xlabel("Year")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```

OUTPUT:


Figure 1
Crop Production Heatmap (State vs Year)
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Chhattisgarh
Delhi
Goa
Gujarat
Haryana
Himachal Pradesh
Jammu and Kashmir
Jharkhand
Karnataka
Kerala
Madhya Pradesh
Maharashtra
Manipur
Meghalaya
Mizoram
Nagaland
Odisha
Puducherry
Punjab
Sikkim
Tamil Nadu
Telangana
Tripura
Uttar Pradesh
Uttarakhand
West Bengal
State
2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015
Year
1e9
6
5
4
3
2
1


Figure 1
Rainfall Heatmap (Subdivision vs Year)
ANDAMAN & NICOBAR ISLANDS
ANDHRA PRADESH
ARUNACHAL PRADESH
ASSAM
BIHAR
CHHATTISGARH
DELHI
GOA
GUJARAT
HIMACHAL PRADESH
JAMMU & KASHMIR
JHARKHAND
KARNATAKA
KERALA
LAKSHADWEEP
MADHYA PRADESH
MAHARASHTRA
NAGALAND
ORISSA
PUNJAB
RAJASTHAN
SIKKIM
TAMIL NADU
TELANGANA
UTTAR PRADESH
UTTARAKHAND
WEST BENGAL
Subdivision
2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015
Year
4000
3500
3000
2500
2000
1500
1000
500
0
(x, y) = (,)

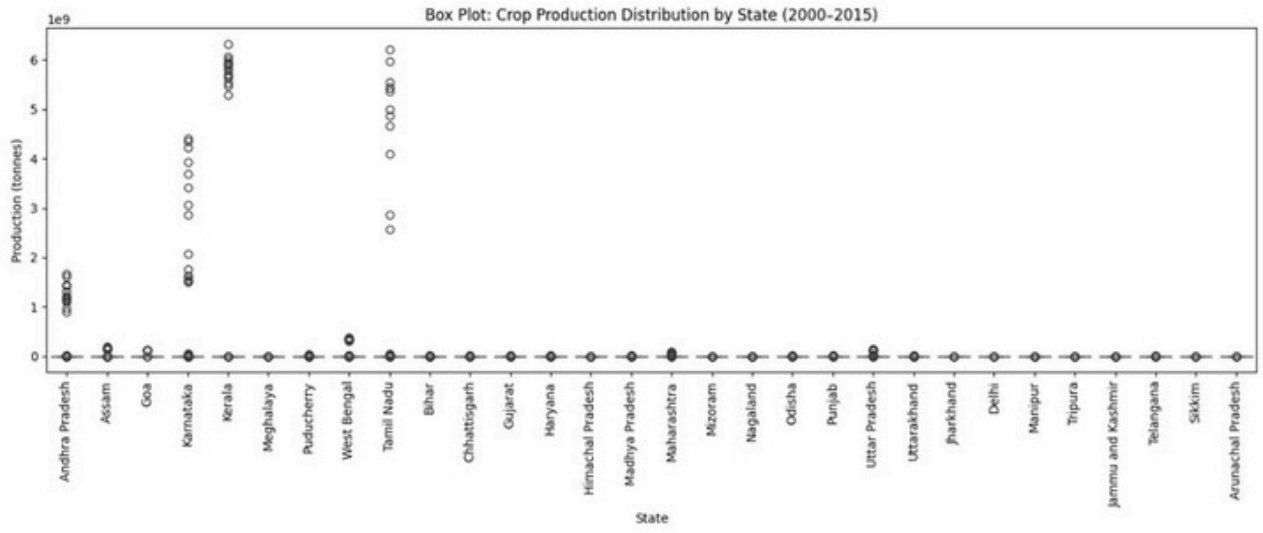
3. **BOX PLOT**

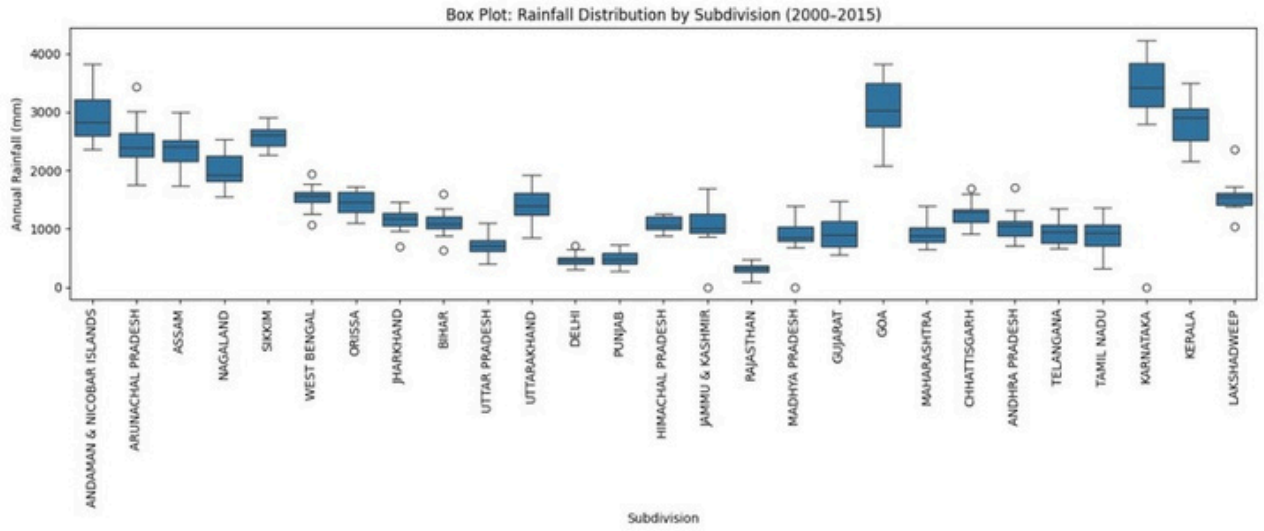
CODE:

```
# --- BOX PLOTS ---
plt.figure(figsize=(14, 6))
sns.boxplot(data=rainfall_df, x="SUBDIVISION", y="ANNUAL")
plt.xticks(rotation=90, fontsize=10)
plt.title("Box Plot: Rainfall Distribution by Subdivision (2000-2015)")
plt.xlabel("Subdivision")
plt.ylabel("Annual Rainfall (mm)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(14, 6))
sns.boxplot(data=crop_df, x="State", y="Production")
plt.xticks(rotation=90, fontsize=10)
plt.title("Box Plot: Crop Production Distribution by State (2000-2015)")
plt.xlabel("State")
plt.ylabel("Production (tonnes)")
plt.tight_layout()
plt.show()
```

OUTPUT:





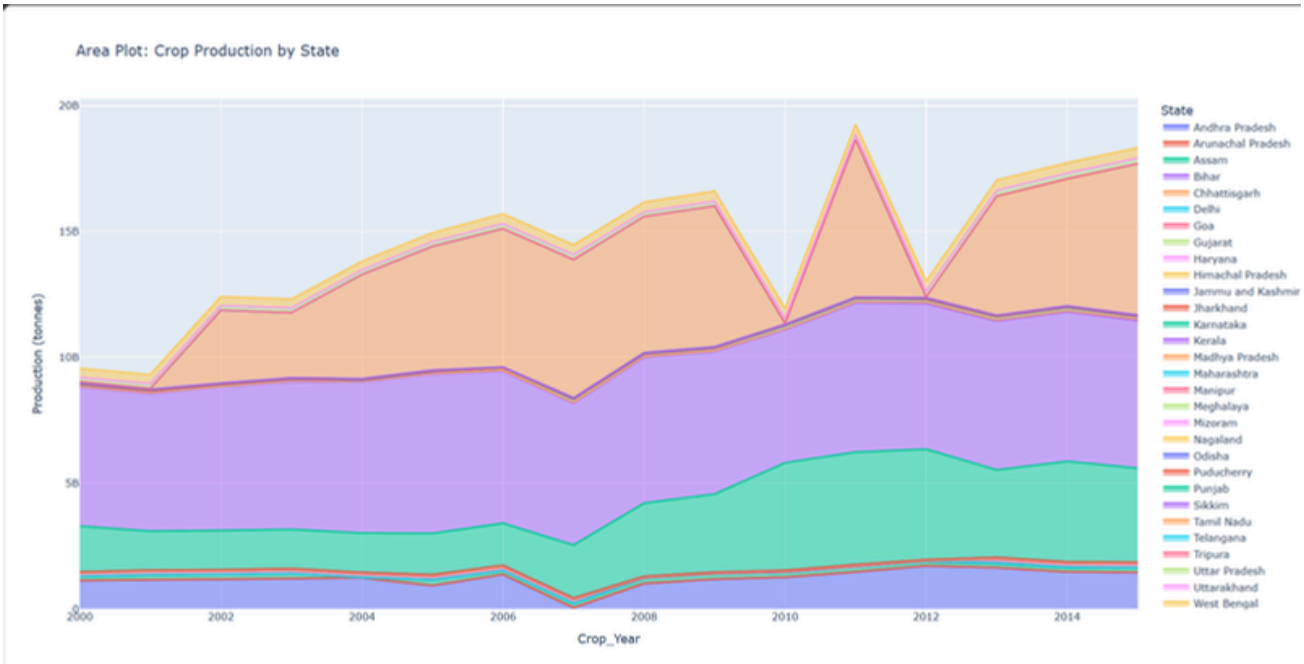
4. **AREA PLOT**

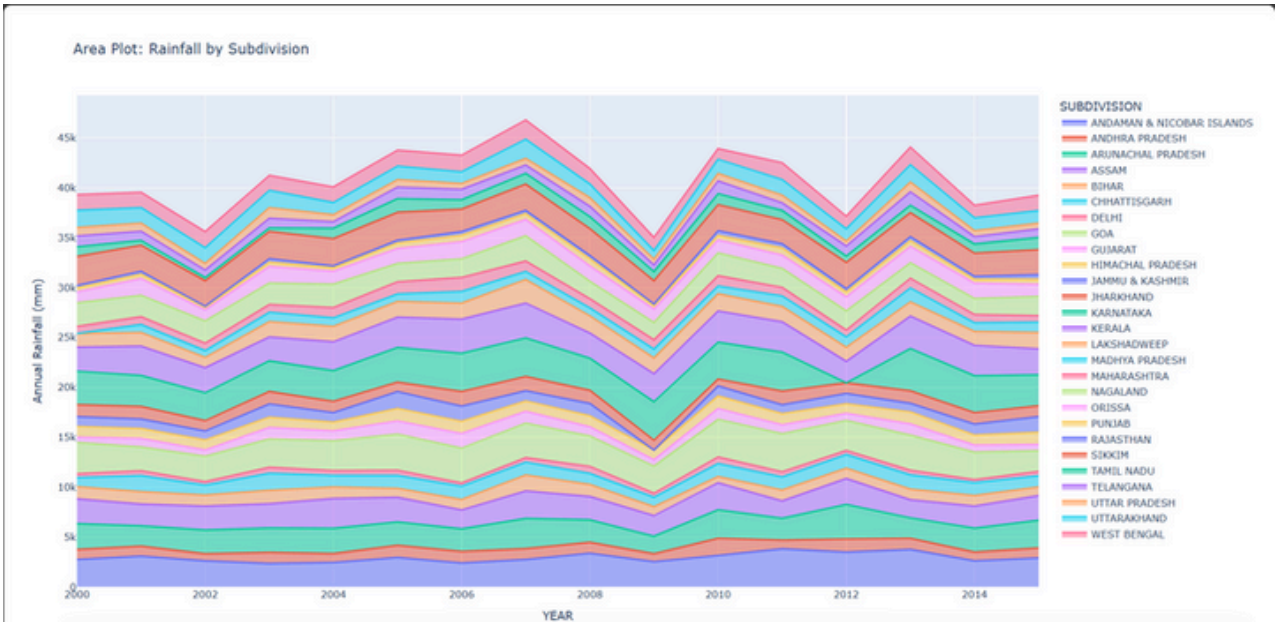
CODE:

```
# --- AREA PLOTS ---
crop_area = crop_df.pivot_table(index="Crop_Year", columns="State", values="Production", aggfunc="sum")
fig_crop_area = px.area(crop_area, title="Area Plot: Crop Production by State", labels={"value": "Production (tonnes)", "variable": "State", "index": "Year"})
fig_crop_area.show()

rainfall_area = rainfall_df.pivot_table(index="YEAR", columns="SUBDIVISION", values="ANNUAL", aggfunc="mean")
fig_rainfall_area = px.area(rainfall_area, title="Area Plot: Rainfall by Subdivision", labels={"value": "Annual Rainfall (mm)", "variable": "Subdivision", "i
fig_rainfall_area.show()
```

OUTPUT:





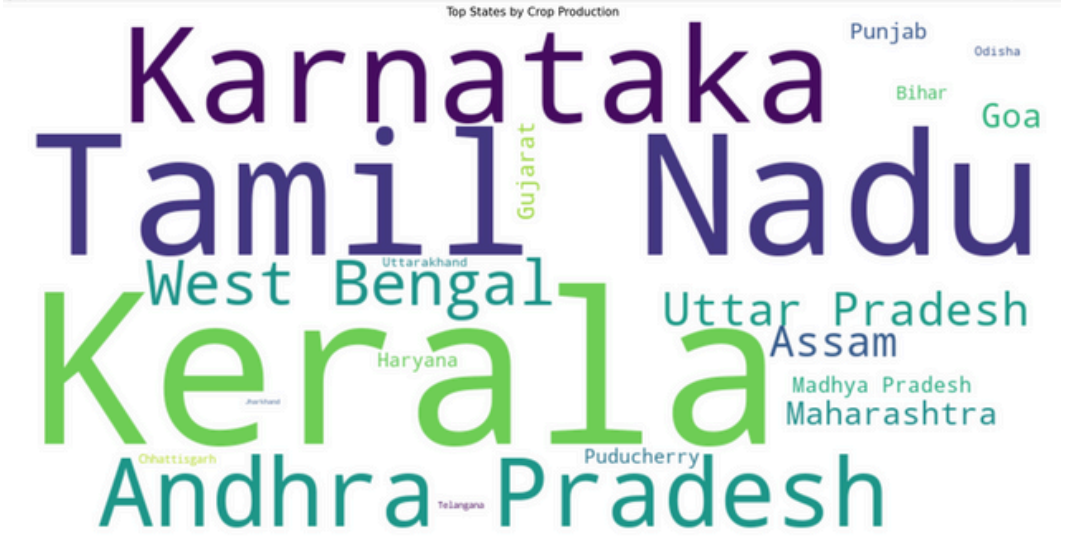
5. **WORD CLOUD**

CODE:

```
# WORD CLOUDS (Top 20)
top_states = crop_df.groupby("State")["Production"].sum().sort_values(ascending=False).head(20).to_dict()
wc_crop = WordCloud(width=800, height=400, background_color='white').generate_from_frequencies(top_states)
plt.figure(figsize=(10, 6))
plt.imshow(wc_crop, interpolation="bilinear")
plt.axis("off")
plt.title("Top States by Crop Production")
plt.tight_layout()
plt.show()

top_subdivs = rainfall_df.groupby("SUBDIVISION")["ANNUAL"].mean().sort_values(ascending=False).head(20).to_dict()
wc_rain = WordCloud(width=800, height=400, background_color='white').generate_from_frequencies(top_subdivs)
plt.figure(figsize=(10, 6))
plt.imshow(wc_rain, interpolation="bilinear")
plt.axis("off")
plt.title("Top Subdivisions by Average Rainfall")
plt.tight_layout()
plt.show()
```

OUTPUT:





OVERVIEW AND INSIGHTS OF THE VISUALIZATIONS

DATASET OVERVIEW

Two datasets were used:

- Crop Yield Dataset: Contains production data by state and year.
- Rainfall Dataset: Contains annual rainfall data across various subdivisions in India.

Missing values were filled with zeros, and the datasets were filtered to focus on the 2000–2015 timeframe for consistency and relevance.

VISUALIZATION SUMMARY AND INSIGHTS

1. **Word Clouds**

- Crop Production by State: Displays the top 20 states by total crop production. Larger font size represents higher production.
- Average Rainfall by Subdivision: Highlights subdivisions with the highest average annual rainfall.

INSIGHT:

States like Punjab, Uttar Pradesh, and Maharashtra prominently feature in crop production, while regions in Northeast and Coastal India lead in rainfall.

2. **Bar Plots**

Crop Production by Year and State: Compares yearly crop production for each state.

Rainfall by Year and Subdivision: Visualizes how rainfall trends vary over time across different regions.

INSIGHT:

Production is generally steady with occasional spikes, possibly due to better monsoons or policies. Rainfall patterns are more erratic, indicating climatic variation.

3. **Heatmaps**

Rainfall Heatmap (Subdivision vs Year): Depicts rainfall intensity over the years across subdivisions.

Crop Production Heatmap (State vs Year): Shows production distribution trends over time.

INSIGHT:

Certain subdivisions show declining or fluctuating rainfall, which may correlate with shifts in crop productivity.

4. **Box Plots**

Rainfall Distribution: Visual comparison of rainfall variability within each subdivision.

Crop Production Distribution: Box plots for states illustrate variation and outliers in production levels.

INSIGHT:

Significant disparity exists in rainfall and crop production, with some states/subdivisions showing high variability and others more stability.

6. **Area Plots**

Production by State (Cumulative): Visualizes the aggregated contribution of each state over time.

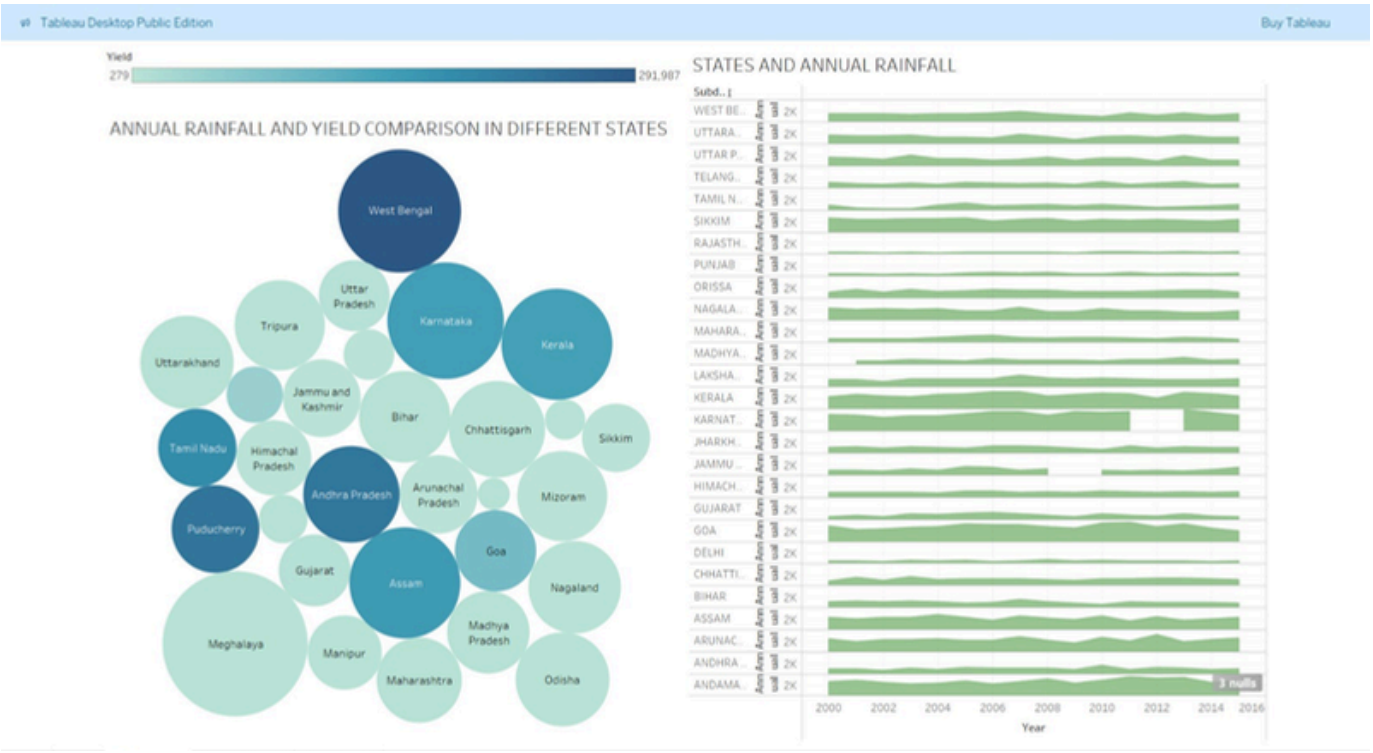
Rainfall by Subdivision: Tracks average annual rainfall contribution visually.

INSIGHT:

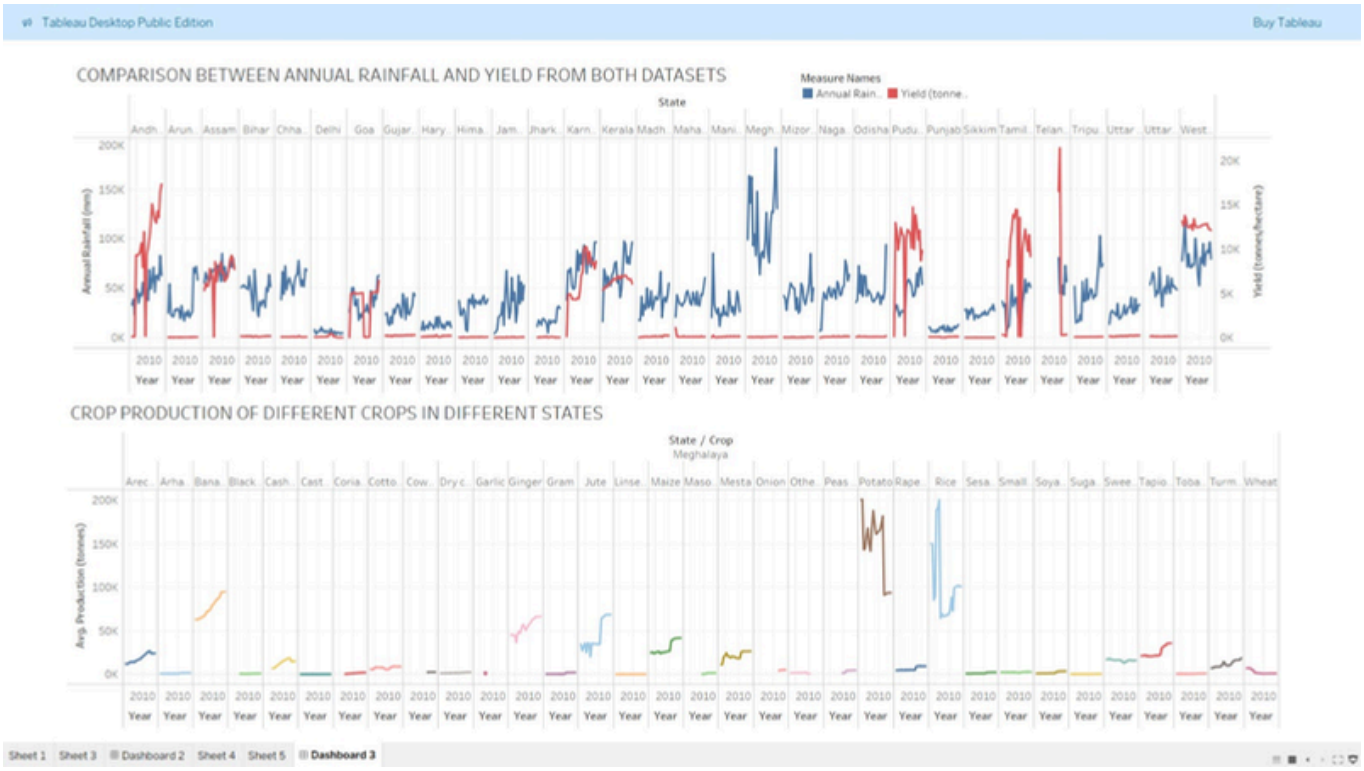
Helps in understanding regional dominance and comparative analysis over time.

TABLEAU DASHBOARD

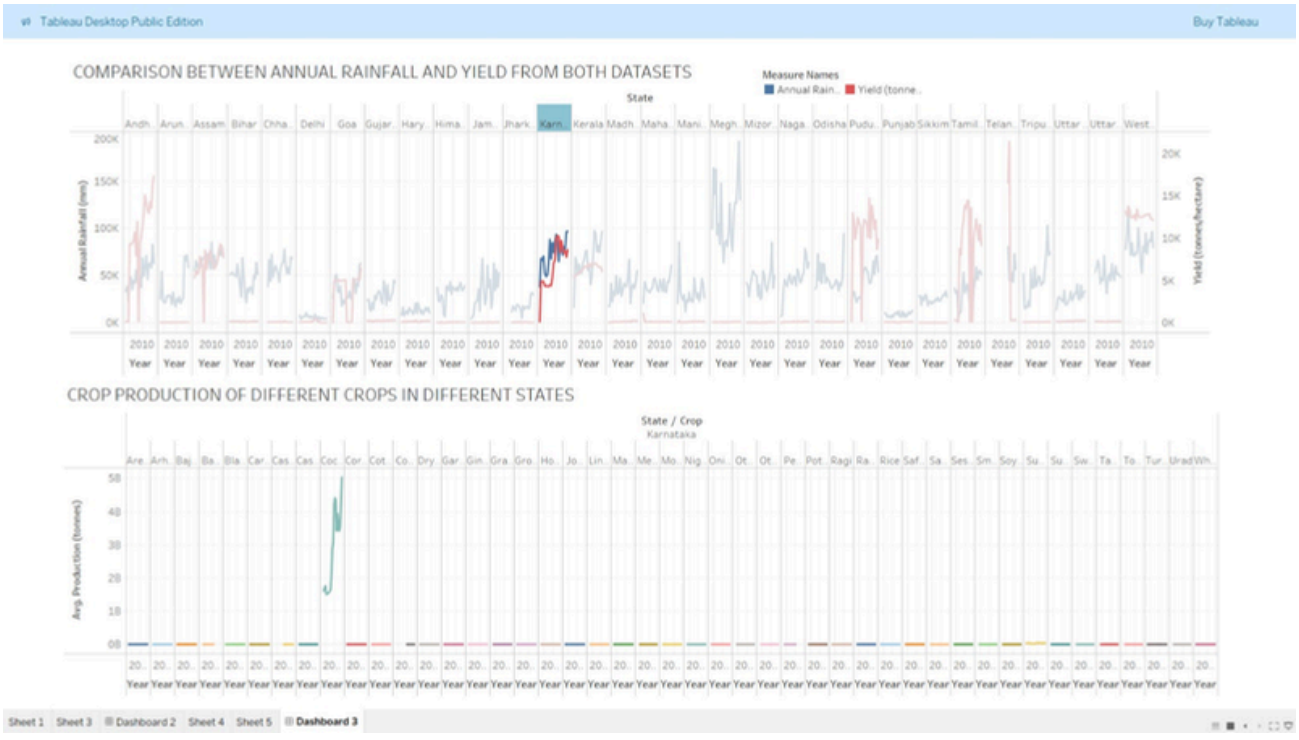
Dashboard visualizing crop yield dataset(left) and rainfall dataset (right individually)

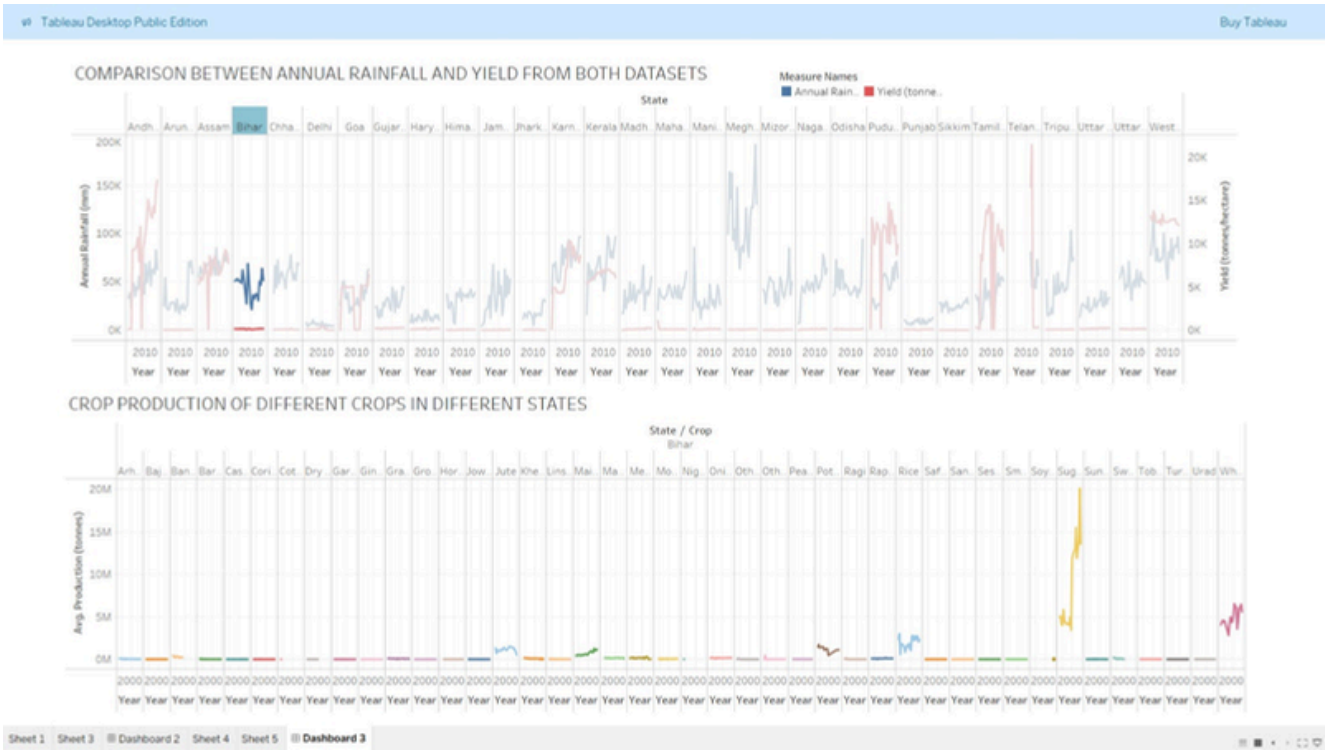


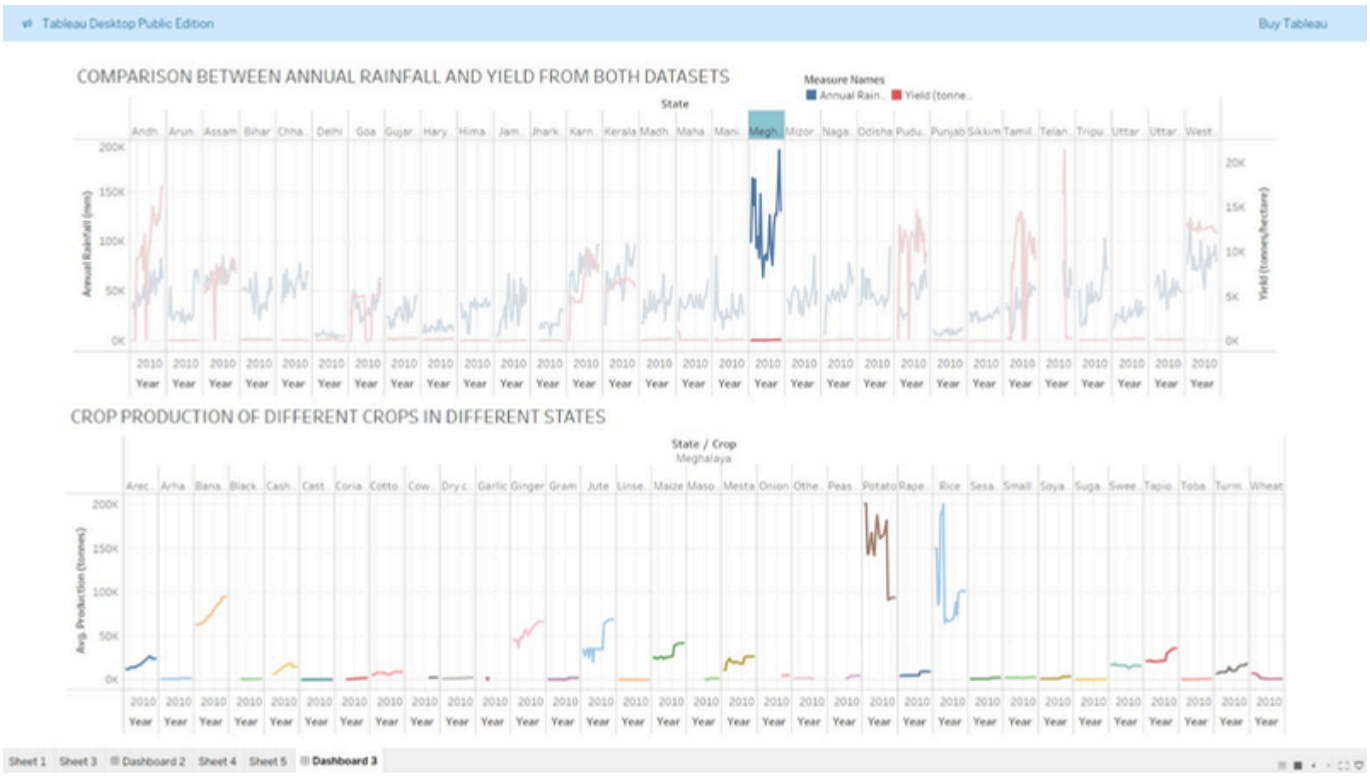
This visual dashboard presents a comparative analysis of crop yield and rainfall across different states. It simultaneously illustrates the amount of
rainfall received and the corresponding crop yield produced in each state, based on data from a merged dataset.

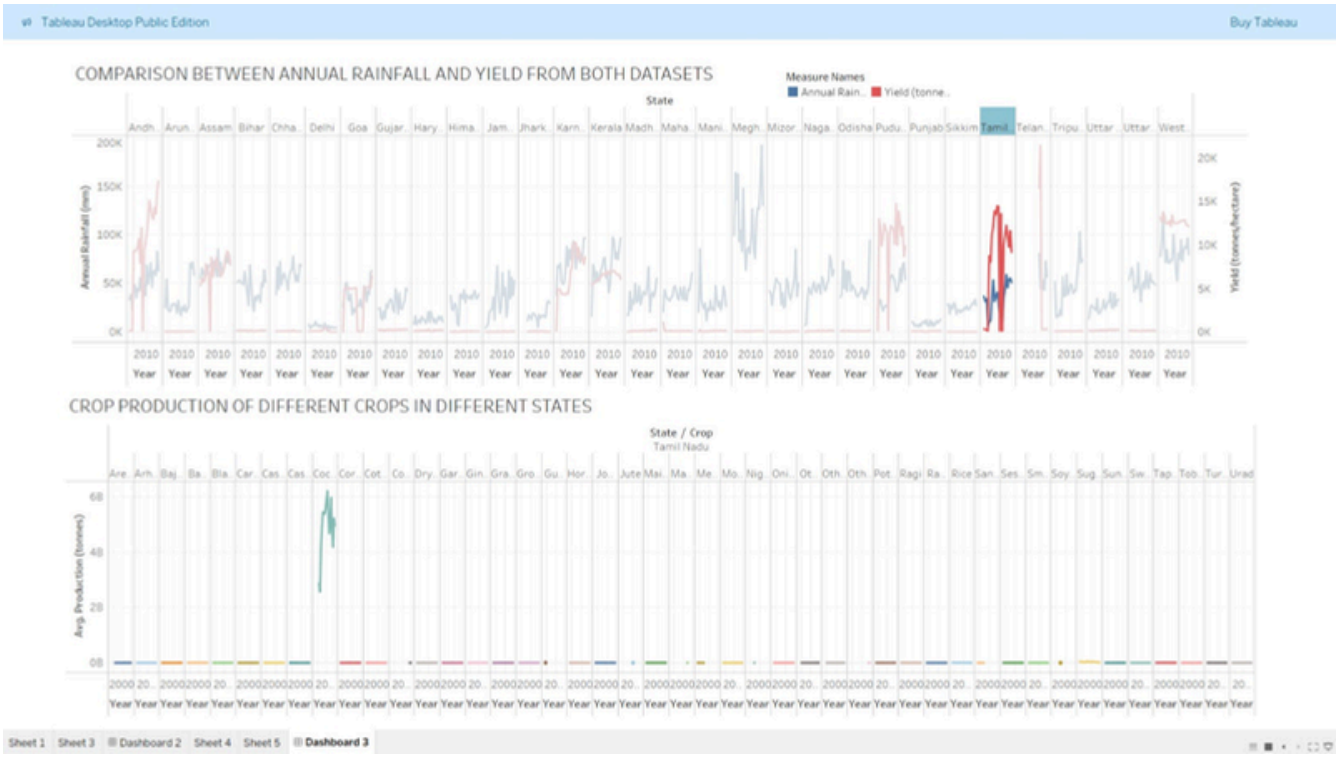


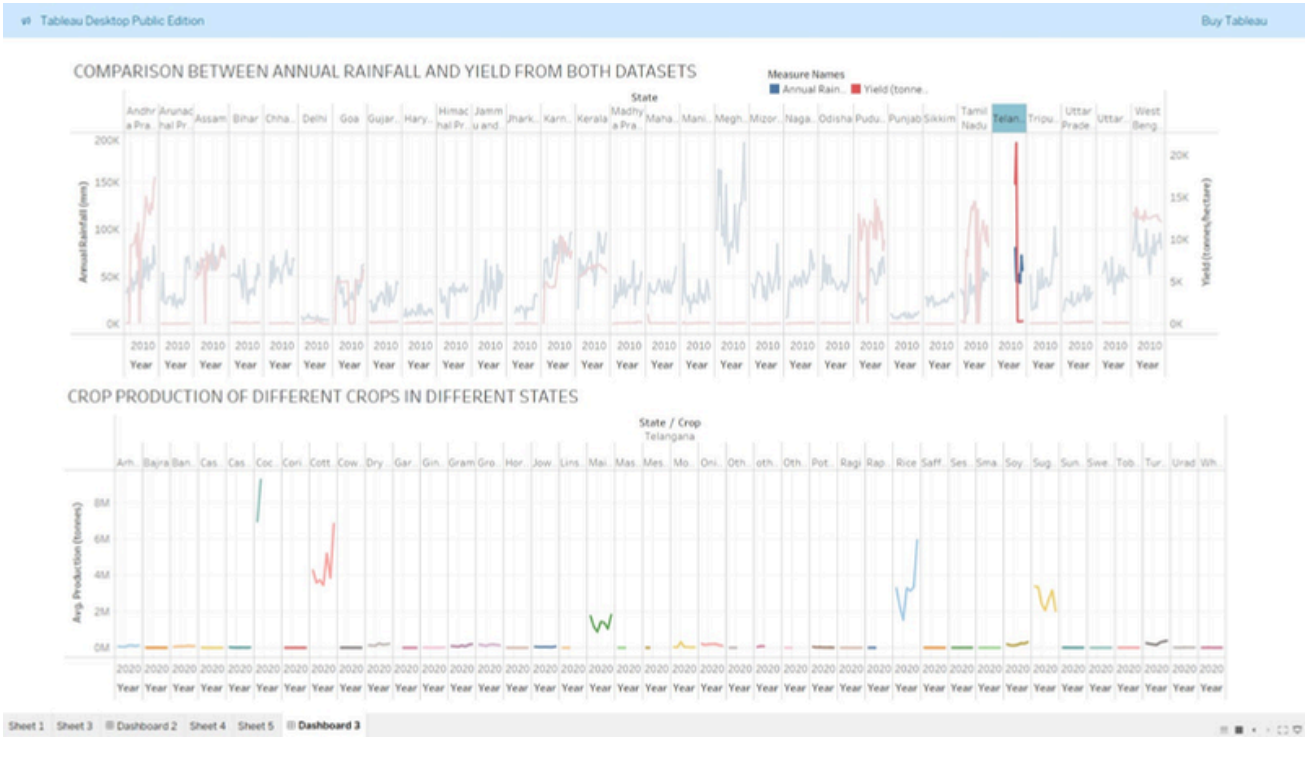
Let us take a look at the visual comparisons of different states as presented in the dashboard.




Tableau Desktop Public Edition
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COMPARISON BETWEEN ANNUAL RAINFALL AND YIELD FROM BOTH DATASETS
Measure Names
Annual Rain..
Yield (tonne..
State
Andh.. Arun.. Assam Bihar Chha.. Delhi Goa Gujar.. Hary.. Hima.. Jam.. Jhark.. Karn.. Kerala Madh.. Maha.. Mani.. Megh.. Mizor.. Naga.. Odisha Pudu.. Punjab Sikkim Tamil.. Telan.. Tripu.. Uttar.. Uttar.. West..
Annual Rainfall (mm)
Yield (tonnes/hectare)
Year
CROP PRODUCTION OF DIFFERENT CROPS IN DIFFERENT STATES
State / Crop
Bihar
Arh.. Baj.. Ban.. Bar.. Cas.. Cori.. Cot.. Dry.. Gar.. Gin.. Gra.. Gro.. Hor.. Jow.. Jute Khe.. Lins.. Mai.. Ma.. Me.. Mo.. Nig.. Oni.. Oth.. Oth.. Pea.. Pot.. Ragi Rap.. Rice Saf.. San.. Ses.. Sm.. Soy.. Sug.. Sun.. Sw.. Tob.. Tur.. Urad Wh..
Avg. Production (tonnes)
Sheet 1
Sheet 3
Dashboard 2
Sheet 4
Sheet 5
Dashboard 3


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COMPARISON BETWEEN ANNUAL RAINFALL AND YIELD FROM BOTH DATASETS
Measure Names
Annual Rain..
Yield (tonne..
State
Andh.. Arun.. Assam Bihar Chha.. Delhi Goa Gujar.. Hary.. Hima.. Jam.. Jhark.. Karn.. Kerala Madh.. Maha.. Mani.. Megh.. Mizor.. Naga.. Odisha Pudu.. Punjab Sikkim Tamil.. Telan.. Tripu.. Uttar.. Uttar.. West..
Annual Rainfall (mm)
Yield (tonnes/hectare)
Year
CROP PRODUCTION OF DIFFERENT CROPS IN DIFFERENT STATES
State / Crop
Meghalaya
Arec.. Arha.. Bana.. Black.. Cash.. Cast.. Coria.. Cotto.. Cow.. Dry c.. Garlic Ginger Gram.. Jute Linse.. Maize Maso.. Mesta Onion Othe.. Peas.. Potato Rape.. Rice Sesa.. Small.. Soya.. Suga.. Swee.. Tapio.. Toba.. Turm.. Wheat
Avg. Production (tonnes)
Sheet 1
Sheet 3
Dashboard 2
Sheet 4
Sheet 5
Dashboard 3


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COMPARISON BETWEEN ANNUAL RAINFALL AND YIELD FROM BOTH DATASETS
Measure Names
Annual Rain..
Yield (tonne..
State
Annual Rainfall (mm)
Yield (tonnes/hectare)
Year
CROP PRODUCTION OF DIFFERENT CROPS IN DIFFERENT STATES
State / Crop
Tamil Nadu
Avg. Production (tonnes)
Sheet 1
Sheet 3
Dashboard 2
Sheet 4
Sheet 5
Dashboard 3


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COMPARISON BETWEEN ANNUAL RAINFALL AND YIELD FROM BOTH DATASETS
Measure Names
Annual Rain..
Yield (tonne..
State
Annual Rainfall (mm)
Yield (tonnes/hectare)
Year
CROP PRODUCTION OF DIFFERENT CROPS IN DIFFERENT STATES
State / Crop
Telangana
Avg. Production (tonnes)
Sheet 1
Sheet 3
Dashboard 2
Sheet 4
Sheet 5
Dashboard 3

Visualization on Folium

CODE:

```
import json
import folium

# Loading summarized crop and rainfall data
with open("state_prod_rain_fixed1.json", "r") as f:
    state_summary = json.load(f)

# Loading India GeoJSON
with open("india_states.geojson", "r", encoding="utf-8") as f:
    india_geo = json.load(f)

# Normalize GeoJSON features with lowercase state IDs
for feature in india_geo["features"]:
    feature["id"] = feature["properties"]["NAME_1"].strip().lower()

# Preparing data for choropleth
choropleth_data = []
prod_values = []

for state, stats in state_summary.items():
    state_id = state.strip().lower()
    if state_id in [f['id'] for f in india_geo['features']]:
        choropleth_data.append([state_id, stats["Total Production"]])
        prod_values.append(stats["Total Production"])

# Determining threshold for top producers (90th percentile)
prod_threshold = sorted(prod_values)[int(0.9 * len(prod_values))]

# Create Folium map
m = folium.Map(location=[23.5, 80.5], zoom_start=5, tiles="CartoDB positron")

# Adding Choropleth layer
folium.Choropleth(
    geo_data=india_geo,
    data=choropleth_data,
    columns=[0, 1],
    key_on="feature.id",
    fill_color="YlGn",
    fill_opacity=0.7,
```

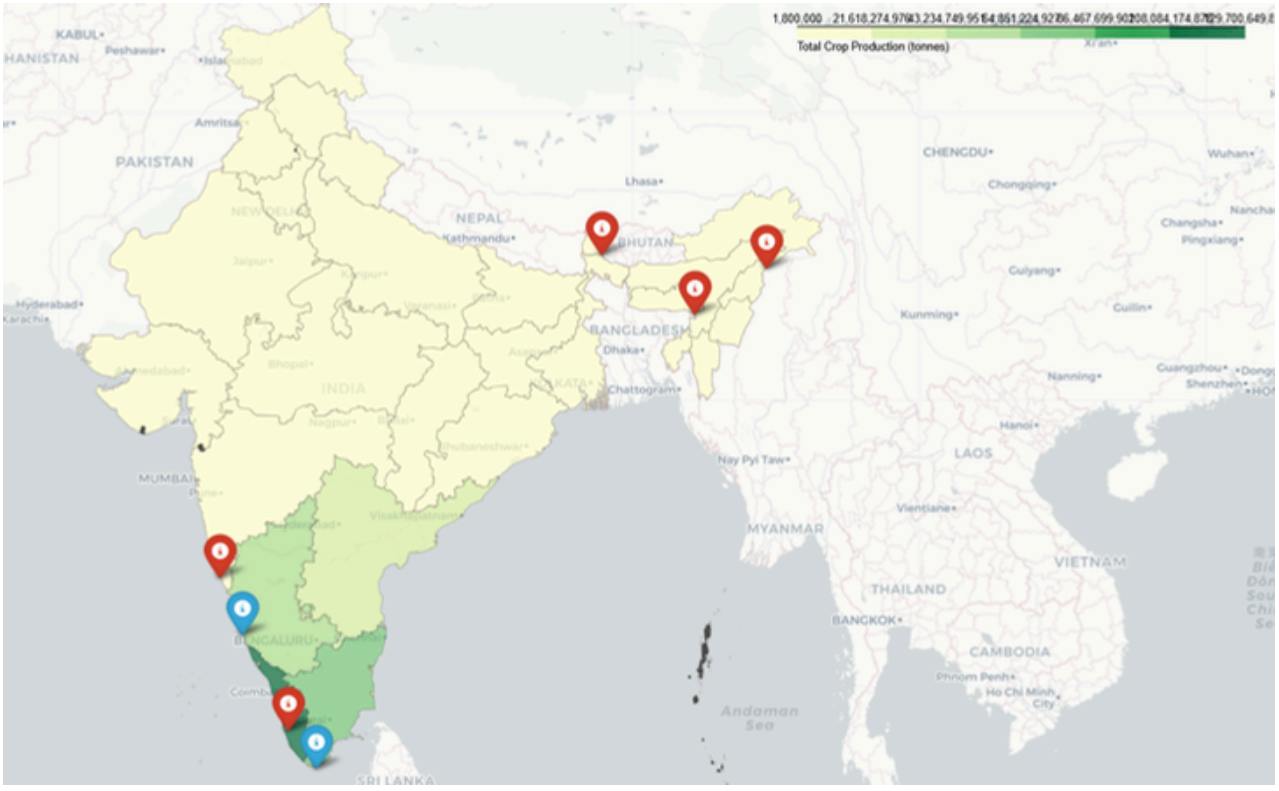
```
    line_opacity=0.2,
    legend_name="Total Crop Production (tonnes)"
).add_to(m)

# Helper function to get centroid for placing markers
def get_centroid(geometry):
    coords = geometry["coordinates"]
    if geometry["type"] == "Polygon":
        lon_lat = coords[0][0]
    elif geometry["type"] == "MultiPolygon":
        lon_lat = coords[0][0][0]
    else:
        return None
    return [lon_lat[1], lon_lat[0]]  # Return [lat, lon]

# Add markers for high rainfall or high production states
for feature in india_geo["features"]:
    state_id = feature["id"]
    stats = state_summary.get(state_id.title()) or state_summary.get(state_id)
    if stats:
        if stats["Avg Rainfall"] > 2500 or stats["Total Production"] > prod_threshold:
            centroid = get_centroid(feature["geometry"])
            if centroid:
                folium.Marker(
                    location=centroid,
                    popup=(f"{feature['properties']['NAME_1']}<br>"
                           f"Rainfall: {stats['Avg Rainfall']:.1f} mm<br>"
                           f"Production: {stats['Total Production']:.0f} tonnes"),
                    icon=folium.Icon(color="red" if stats["Avg Rainfall"] > 2500 else "blue")
                ).add_to(m)

# Save map
m.save("cleaned_folium_map.html")
print("✅ Map saved as 'cleaned_folium_map.html'")
```

OUTPUT:



RED AND BLUE MARKERS (ICON PINS)

These markers indicate states of special interest, based on two key conditions:

🔴**Red Marker:** States where the average rainfall exceeds 2500 mm. Ex. Kerala and Goa, Northeast states

🔵**Blue Marker:** States with total crop production above the 90th percentile (top 10% of producers). Ex. Tamil Nadu, Karnataka.

CHOROPLETH COLOR SHADES (YELLOW TO GREEN)

These shades represent the total crop production per state, derived from the

🟨**Light Yellow** →Lower production

🟩**Darker Green** → Higher production

NOTE:

Because Kerala, Tamil Nadu, and Karnataka have extremely high values (in the tens of billions), the gradient is skewed — so states like Punjab or UP may appear lighter even if they produce a lot.

CONCLUSION

Rainfall remains one of the most influential natural factors in Indian agriculture. Its effects on crop yield and production are significant, especially in rain-dependent regions.

1. Rainfall Drives Kharif Crop Success

- Kharif crops (e.g., rice, maize, pulses) are sown with the onset of the monsoon (June–September).
- Adequate and well-distributed rainfall during this period is crucial for germination, growth, and harvest.
- Shortfall or delay in monsoon affects sowing windows, leading to reduced acreage and lower yield.

2. Rainfall Variation Causes Yield Instability

- States like Maharashtra, Odisha, and Madhya Pradesh showed visible dips in crop production during drought years (e.g., 2002, 2004, 2009).
- These are predominantly rain-fed regions, with limited irrigation infrastructure, making them highly vulnerable.

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3. Temporal Correlation Patterns

- Yearly data show clear temporal links between rainfall dips and production drops.

- For example:

In 2002 and 2009, drought conditions led to substantial national-level yield reductions.

- Rain-fed states saw the most severe declines, especially in non-irrigated zones like Marathwada and Vidarbha (Maharashtra).

4. Uneven Rainfall Affects Regional Productivity

Regional imbalances in rainfall:

- Coastal and North-Eastern states receive excess rainfall, often leading to waterlogging or soil erosion.

- Arid zones (e.g., Rajasthan) receive insufficient rainfall, limiting crop variety and yield potential.

5. Rainfall Dependency Differs by State

- High dependency in rain-fed states: Crop output rises or falls with monsoon performance.

- Moderate dependency in semi-irrigated states: Production dips slightly in drought years but generally remains stable.

- Low dependency in irrigation-intensive states: Yields remain consistent regardless of rainfall variation.

Why Rainfall Is Not the Only Factor in Crop Production?

Although rainfall is critical, it is not the sole determinant of agricultural success. Several non-climatic factors significantly influence crop yield and overall production.

1. Irrigation Infrastructure is Crucial

- States like Punjab and Haryana consistently produce high yields with moderate and stable rainfall.

Their advantage comes from:

- Extensive canal networks
- Tube well irrigation
- Water-intensive cropping patterns (e.g., rice-wheat)

2. Technology and Mechanization

- Use of improved seed varieties, fertilizers, pesticides, and farm machinery boosts productivity even in less favorable climates.
- Mechanization reduces dependence on climate, enabling timely planting and harvesting.

3. Policy and Institutional Support

- Minimum Support Prices (MSPs), government procurement, crop insurance, and input subsidies help reduce climate-related risks.

- These supports are stronger in politically prioritized agricultural states.

4. Crop Selection and Agro-ecological Adaptation

- Inland Karnataka achieves decent yields by growing drought-tolerant crops like millets and pulses, despite low rainfall.

- In contrast, North-East India, despite abundant rain, shows low yields due to unsuitable terrain and crops, and poor logistics.

5. Socioeconomic and Structural Factors

- Access to markets, farmer education, landholding size, and credit availability play vital roles.

- Smallholders without irrigation or financial backup are more exposed to climate risks.